Welcome to this edition of the **University IT notes**.

This is a summarized version of different notes that may appear in year one within the first semester. These notes are useful for all students doing IT as a major, Business students and those who wish to have **basic information** about **Computers** and **Computer practicals.**

For those students who are in need of practicals, this is your go to document. IT favours **BIT, DIT, BBA, DBA, BSAF and IT** students who may be doing education.

This is the first edition of this copy for **@2025/2026**, semester one in year one. Grab your copy before exams find you.

List of Course units.

1. **Computer literacy notes** and practicals in productivity softwares.
2. Introduction to **information systems** and **system modeling.**
3. Introduction to i**nformation technology.**
4. Introduction to **website designing.**

This document will be so useful to students who want to **pass the theory in exams** and those who want to do the **practical parts and want to master the basics** like how to use a computer, **computer shortcuts** and **personal training** from the auhor of this document.

Get your copy as early as possible and start practicing, this will remove the fear of failure as it combines most essentials for this semester one for year ones.

Other copies.

1. Year two summaries and programming languages like **java, python and c++.**
2. Year three summaries and notes including **advanced programming**.

Note:

This is a **free copy,** the premium copy has more practical sessions, exam copies and answers, Deep insights and clickable links for searching different data. The premium copy also has no watermarks meaning you can buy it and re-create your own copies for sale.

This information is to help students address computer issues from a basic level, **the medium and advanced levels are sold @ 5k UGX,** online payments, no hardcopy creations.

mail to: **ashrafsoftwareguy@gmail.com** or whatsapp **0727362961**

Semester Summary for Year One
Semester one

Computer

This is an electronic device that receives **input, processes** it using instructions, **stores** that data but produces it as information **output.** It operates under the control of programs and can perform arithmetic, logical and data manipulation tasks with speed and accuracy.

History and Evolution of Computers

1. **Early Foundations -** It had the **Abacus** (circa 2400 BC) which was the first known mechanical calculator. Charles Babbage in 1830s came in with the **Analytical Engine**, considered the first concept of a programmable computer with Ada Lovelace as the first programmer.
2. **Mechanical to Electronic shift** - In 1940s transition from mechanical machines to elctronic computers began, because thats when the stored program concept also had initiation.

Generations of computers

1. **First generation (1940-1956)**. This generation was characterized of using vacuum tubes for circuitry and magnetic drums for memory. They used punched cards for input and output, **machine language** in binary, had examples like **ENIAC and UNIVAC.** These computers where **large, expensive, high heat output** due to vacuum tubes and **slow processing** power.
2. **Second generation (1956-1963).** In this shift **transistors** replaced vacuum tubes and **assembly language** was introduced. examples had **IBM 1401, CDC 1604.** Some of the features where that these computers where **smaller, faster, more reliable and had less heat** compared to those in the firta generation.
3. **Third generation (1964).** Introduction of **intergrated circuits (ICs)** was seen as a shift in technological advancement. This is where we see introduction of **high level languages** like **FORTRAN, COBOL.** Examples included **IBM system/360.** Features they had where **more compact, multitasking and lower costs.**
4. **Fourth generation (1971-2010).** This generation saw a big advancement int tech where **microprocessors** integrated **CPU** on a single chip. It was **GUI** based with advanced high level programming languages. Examples included **Intel 4004, Apple Macintosh, IBM PC.** Features they had included **PCs, networking, graphical interfaces.**

5. **Fifth generation (2010-Present).** Introduction of **Artificial Intelligence** was realised in this age. They use technologies like **AI, machine learning, quantum computing, cloud systems.** Different languages like **Neural networks, Natural language processing.** Examples include **IBM Watson, Google DeepMind, Modern Smartphones.** Features include **Intelligent systems, voice recognition, automation.**

Characteristics of computers

1. **Speed.** Computers can perform millions or billions of instructions per second. Super computers can process trillions of calculations per second.
2. **Accuracy.** They are capable of performing different tasks with minimal errors provided input and instructions are correct.
3. **Automation.** Once programmed, computers execute tasks without human intervention. Like automated billing systems or industrial robots.
4. **Storage.** Computers can store vast amounts of data and retrieve it instantly for example cloud servers that store TBs of user data.
5. **Versatility.** They can perform a variety of tasks for example they can switch from gaming to data analysis with ease, or gaming back to the browser without first changing the computer.
6. **Diligence.** Unlike humans, computers dont suffer from fatigue or boredom or tiredness. For example running 24/7 security monitoring systems without performance drop.
7. **Mulitasking.** Capable of handling multiple applications simultaneously. They can stream, download and edit at once.
8. **Artificial Intelligence.** They are used to simulate human like thinking and can answer according to the vectors given and scope.

Classification of computers

1. **Size and Capacity.** These include super computers, mainframes, minicomputers, microcomputers and embedded systems.
2. **Processing power.** They include analog , digital and hybrid computers.
3. **Purpose.** They range from general purpose to special purpose computers.
4. **Functionality.** These computers include wearables, mobile devices, workstations and servers.

Computer hardware

It is the collection of physical devices that make up a computer system. It includes everything you can touch on on a computer.

Types of Computer Hardware

1. **Input Devices.** They are used to feed data into the computer like keyboards, mouse, scanners, microphones.
2. **Output Devices.** They display or disseminate results from the computer like monitors, speakers, printers, plotters.
3. **Processing Devices.** These perform calculations and logical operations within the computer like CPU.
4. **Storage Devices.** They are used to save data temporarily or permanently like RAM, HDD, SSDs.
5. **Communication Devices.** These enable exchange of data between systems. like NICs, modems, routers.

Central Processing Unit (CPU)

It executes instructions from the software and coordinates all hardware operations.

CPU cycle - It includes the **Fetch** which retrieves instructions and data from memory or registers, **Decode** which interpretes instructions from fetch, **Execute** that performs the operations like calculations, data movement etc, **Store** which stores results from processes.

Types of Processors

1. **CPU.** This is the Central Processing Unit, a general purpose processor for all computing tasks like Intel core i7, AMD Ryzen 9.
2. **GPU.** Graphical Processing Unit, it is optimized for parallel processing and rendering graphics like NVIDIA RTX 4090, AMD Radeon RX 7900.
3. **TPU.** Tensor Processing Unit, designed by Google for machine learning and AI workloads like Google TPU v4 used in deep learning models.
4. **QPU.** Quantum Processing Unit, used in quantum computers to process quantum bits (qubits) like IBM Q system One, D-Wave Advantage.
5. **NPU.** Neural Processing Unit, it is specialized for AI inference tasks in mobile and edge devices like Apple Neural Engine, Huawei Kirin NPU.

6. **FPGA.** Field-Programmable Gate Array, reconfigable chip used in custom hardware application like Xilinx Virtex series.
7. **DPU,** Data Processing Unit, optimized for data movement and networking tasks like NVIDIA BlueField DPU.

Storage Devices

1. **Primary Storage.** This is also referred to as the main memory, it includes **RAM** which is a temporary fast memory for active tasks, **ROM** which stores firmware and boot instructions.
2. **Secondary Storage.** These keep data permanently for future usage like the HDDs, SSDs, optical drives, flash disks.

Computer Software

These are a set of instructions and programs that tell a computer what to do and how to do specific tasks. They enable hardware to function and perform tasks.

Types of Computer Software

1. **System software.** This controls and manages hardware and basic system operations which acts as a platform for running application software. Examples include the OS like Windows, MacOS, Android, iOS, Ubuntu. Utilities like disk cleanup, antiviruses, file management tools. Device drivers like printer drivers, graphics card drivers. Firmware like BIOS, embedded system firmware, UEFI. Language translators like compilers, interpreters, assemblers, linkers. Shells like CMD, Bash, Zsh, Csh.
2. **Application software.** This is designed for users to perform specific tasks and runs on top of system software. Examples include productivity tools, media and design software, web and communication software, education and coding software.

Operating System

An OS is the most critical system software that acts as an interface between the user and hardware, managing resources and providing services for applications. Examples of OSes include, Desktop OS, Mobile OS, Server OS and Embedded OS.

System Software VS Application Software

1. **Purpose,** System software manages hardware and system operations but application software performs specific user tasks.
2. **Interaction,** System software interfaces with hardware yet application software interfaces with users.

3. **Installation,** System software comes pre-installed or with hardware yet application software is installed by the user.
4. **Dependency,** System software is required for the system to run yet application software depends on system software to run.

File Management System

A file management system is software that enables users to create, organize, access and manage files and folders on a computer. Examples of file managers include Windows File Explorer, macOS Finder, Linux Nautilus.

Basic File Operations

1. **Creating a file.**

   Right click→New→Choose file type(word, text, powerpoint).

2. **Creating a folder.**

   Right click→New Folder→Name it anything you want.

3. **Saving files in folders.**

   Goto File→Save As→choose location and name→insert the correct file format or extension.

4. **Renaming Files.**

   Right click→Rename→Type new name but avoid special characters.

File Types and Extensions

1. Documents (.docx, .pdf, .txt), we use these in word, pdfs and plain text.
2. Images (.jpg, .png, .svg), we use these for photos, pictures and graphics.
3. Audio (.mp3, .wav), we use these for music audios and recordings.
4. Video (.mp4, .avi), we use these for movies, lectures.
5. Spreadsheet (.xlsx, .csv), we use these in data tables, budgets.
6. Database (.odb), we use these for databases.
7. Presentations (.pptx, .odp), we use these for slideshows.
8. Executables (.exe, .app), we use for programs and apps.

Note: Extensions help the system to know which app to use to open the file.

External and Cloud Storage

1. **External Storage.** These are storage devices outside the computer or even portable we use for backups and transferring data. Examples include USB drives, external hard disks, SD cards.
2. **Cloud Storage.** These allow users to access their files form anywhere, auto-sync and share them as long they are connected to the internet. Examples include Google Drive, OneDrive, Dropbox, TeraBox.

Web Browsers

They are applications that allow users to access and interact with websites and web content. Examples of web browsers include, Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera.

Search Engines

These are online tools that index and retrieve information from the web based on user queries. Examples Google, Bing, DuckDuckGo, Yahoo, Yandex.

Safe Browsing Practices

1. Use HTTPS websites
2. Avoid clicking suspicious links or pop-ups
3. Keep your browser and antivirus updated
4. Use strong, unique passwords
5. Avoid public WI-FI for sensitive tasks
6. Enable browser security settings
7. Use ad Blockers and privacy extensions.

Email

It is an electronic mail, digital communication method used to send messages, files and documents over the internet.

Setting Up and Managing Email Accounts

1. Choose a provider like Gmail, Outlook, Yahoo, ProtonMail
2. Visit the provider's website and click Sign Up
3. Enter personal details and choose a secure password

4. Verify your identity via phone or alternative email.

Productivity Software

It refers to applications designed to help users perform tasks such as writing ,calculations, presenting, organizing and collaboration. These tools are essential for students, professionals and businesses. Examples include, Word processors like ms word, google docs. Spreadsheets like ms excel , google sheets. Presentations like ms PowerPoint, google slides. Databases like ms Access. Collaboration softwares like Outlook, Teams, Gmail, Meet, Chat, Drive.

Microsoft Office Tools (Usage)

1. **MS Word**

   Goto Start→type word→click enter→Edit your document→Ctrl+S(save)/ Ctrl+shift+S(Save As) type your name and extension .docx or export as pdf. You can use formatting like headings, set margins, line spacing, apply fonts, colors, borders and themes, use templates, mail merge or insert smart art and images.

2. **MS PowerPoint**

   Goto Start→type powerpoint→click enter→edit→Save and type your name and extension .pptx or export as pdf. You can use format, use slide layouts, tarnsitions, animations, customize fonts, backgrounds, icons, use designer, embed media, record narration.

3. **MS Excel**

   Goto Start→type excel→Enter→edit→Save with any name and extension .xlsx or export as csv or pdf. Excel docs can have adjust cell size, use number formats, conditional formatting, apply themes, charts and formulas, use pivot tables, data validation, macros.

4. **MS Access**

   Goto Start→type excel→Enter→edit→Save with any name and extension .accdb. You can define fields, data types, relationships, use forms, reports and queries for user-friendly views, import or export data, macros.

Google WorkSpace

- **Google Docs.** These enable real-time editing and commenting, auto-save to google drive, share with permissions, use templates, voice typing and version history.
- **Google Sheets.** It is collaborative spreadsheets with formulas and charts, filters, conditional formatting, pivot tables, integrate with forms for data collection, export to excel or pdf.

- **Google Slides.** These create presentations with themes and animations, collaborate and presnt remotely via Meet, embed videos, charts and links, export as .pptx or pdf.

Cybersecurity

It is the art of protecting networks, devices and data from unauthorized access or criminal use, while ensuring confidentiality, integrity and availability of information.

Cybersecurity Basics

1. **Authentication.** Its the use of strong passwords, biometrics or 2FA.
2. **Encryption.** Securing data so that only authorized users can read it.
3. **Firewalls.** Thess block unauthorized access to networks or nodes
4. **Antivirus Software.** It detects and removes malware from computer systems,
5. **Updates and Patches.** These keep current systems upto date and fix vulnerabilities.
6. **User Awareness.** Educating users about phishing , scams and safe practices.

Computer Threats

- Malware like viruses, worms, trojan horses, ransomware
- Phishing attacks like fake emails, site trackers, email scams
- Spyware like software secretly monitoring your computer activity, like key loggers
- Denial of Services or Distributed DOS, overloading systems to make them crash, website outages
- Man-In-The-Middle attacks, they intercept communication between two parties, Wi-Fi eavesdropping

Prevention Measures

- Strong passwords and change them regularly
- Avoid clicking suspicious links
- Install Antivirus and Firewalls
- Backup data regularly
- Use secure networks
- Avoid Public Networks
- Use password hashing

- Always encrypt sensitive information.

Information Security (InfoSec)

It is a broader discipline that focuses on protecting all forms of information using the CIA triad preventing the DAD triad. Cybersecurity is a subset of InfoSec.

Responsible use of Technology

- Respect privacy and data protection laws
- Avoid cyberbullying, hacking or spreading misinformation
- Use devices ethically and avoid illegal downloads or software
- Report suspicious activity or security breaches
- Protection of intellectual property
- Avoid plagiarism

Booting a Computer

It is the process of starting a computer and loading the operating system.

Types of booting

1. Cold booting. This is the starting of a computer from a powered-off state.
2. Warm booting. It is restarting a computer without turning off the power. (Ctrl+Alt+Del)

Shutting Down a computer

Use the start menu→Power→Shutdown.

This ensures all programs close safely and prevents data loss or corruption.

Desktop

The Desktop is the main screen interface of a computer after booting. It displays icons, shortcuts and the taskbar.

Features of a GUI Desktop

- Icons→They represent files, folders and apps.
- Taskbar→It displays open programs, system tray and clock/date.
- Start Menu→Gives access to programs, settings and power options.

- Wallpaper→These are background images or colors.

Navigating the Desktop and File Explorer

1. Desktop navigation→click icons to open apps or files or folders →press Enter/double Enter. Right click for context menu.
2. File Explorer→You can access it via Start→File Explorer or shortcuts, browse drives, folders and files, use the search bar to find items.

A Window

It is a rectangular interface used to display applications or folders.

Components of a Window

1. Title Bar→Shows the name of the app or file
2. Control Buttons→Minimize, maximize, restore or close the window
3. Menu Bar→Contains options like File, Edit, View
4. Scroll Bar→Navigate content vertically or horizontally
5. Content Area→Displays the actual file or app interface

Installing and Uninstalling Software in a computer

1. **Installing.** Download installer→double click→follow setup wizard→choose install location→preferences→confirm installation→launch application.
2. **Uninstalling.** Goto Settings→apps→installed apps→select app→click uninstall→follow prompts to remove completely.

Troubleshooting Basic Computer Issues

- **Software Issues.**

  Slow perfomance→clear cache→close background apps.

  App crashes→reinstall or update the app.

  System error→Use system Restore or Safe Mode

- **Hardware Issues**

  No display→check monitor cables and power.

  Overheating→clean fans, improve ventilation

Peripheral failures→Reconnect or update drivers

Note: Use tools like Event viewer, Task manager, Device Manager to diagnose problems.

Introduction to Coding

Coding or programming is the process of writing instructions that a computer can understand and execute.

Basics of Coding

- Syntax→Rules that define how code is written.
- Variables→These store data (eg name = "Ashiraf")
- Loops→Repeat actions (eg, for and while loops)
- Conditionals→Make decision (if, else)
- Functions→Reusable blocks of code.

Scratch

Is a visual programming language for beginners. It uses drag and drop blocks to create games, animations and stories. It requires no typing, great for learning logic, sequencing, creative thinking, used by kids, educators and beginners worldwide.

HTML

It is HyperTextMarkupLanguage is the standard language for creating web pages. It defines the structure of the web pages like headings, paragraphs, links, images etc. It is often paired with CSS for styling and JavaScript for interactivity.

Essential Keyboard Shortcuts

- Ctrl + C→Copy
- Ctrl + V→Paste
- Ctrl + Z→Undo
- Ctrl + S→Save
- Ctrl + P→Print
- Alt + Tab→Switch apps
- Windows + D→Show Desktop
- Windows + E→Open File Explorer

- Windows + L→Lock screen

Typing Techniques

1. Use all fingers
2. Home row positioning
3. Practice daily
4. Take typing tests
5. Use ergonomic postur
6. Avoid pecking
7. Use a good keyboard

Information Technology

It is the discipline focused on design, implementation, management and support of computer sytems and networks that store, process and transmit data.

Key Functions of IT

- Network administration and cybersecurity
- Database and server management
- Technical support and troubleshooting
- Software deployment and maintenance
- Cloud computing and virtualization

Role of IT in Modern Society

1. Education
2. Healthcare
3. Business
4. Government
5. Communications
6. Finance

Summary

IT is about implementation and support of tech systems.

CS is about creating and understanding the tech itself

GhostR

IS is using tech strategically within organizations

Computer Networks

It is a group of two or more devices connected to share data, resouces and services. Networks can be wired (Ethernet or Fiber) or wireless (radio waves or satellites).

Types of Networks

1. LAN (Local Area Network). This covers a small area like a home, office or school like a computer lab.
2. WAN (Wide Area Network). Covers large geographical areas like the internet
3. MAN (Metropolitan Area Network). Spans a city or campus like city-wide WI-FI
4. PAN (Personal Area Network). Connects personal devices like bluetooth headsets to phones
5. VPN (Virtual Private Network). Creates a secure tunnel over public networks like remote work access

Internet Fundamentals

Internet is a massive global WAN that connects billions of devices. It uses standardized protocols to ensure smooth communication across diverse systems.

Key Internet Concepts

- IP Address→Unique identifier for each device on a network.
- DNS (Domain Name System)→Translates domain names into IP addresses.
- ISP (Internet Service Provider)→Delivers internet access to users.

Network Protocols

These are rules that govern data transmission across networks.

1. TCP (Transmission Control Protocol)→It is a reliable connection-based communication
2. UDP (User Datagram Protocols)→It is a fast, connectionless communication eg streaming
3. HTTP/HTTPS→Web browsing and secure web access
4. FTP (File Transfer Protocol)→Uploading/downloading files over the internet
5. SMTP/IMAP/POP3→Email sending and receiving
6. IP (Internet Protocol)→Routing and addressing packets

mail to: **ashrafsoftwareguy@gmail.com** or whatsapp **0727362961**

Network Devices

1. Router→Connects networks and directs traffic
2. Switch→Connects devices within a LAN
3. Hub→Basic device for connecting multiple devices
4. Access Point→Extends wireless coverage
5. Modem→Converts digital data for transmission over telephone lines and vice versa
6. Firewalls→Monitors and controls incoming/outgoing traffic for security

Data vs Information

Data→These are raw facts and figures without context eg, 23, Uganda, BIT.

Information→Processed and organized data that is meaningful and useful eg Ashiraf is a BIT student in Uganda.

"Data becomes information when it is interpreted and structured for decision-making"

Data Processing Cycle

1. Data Collection→Gathering raw data from sources
2. Data Input→Entering raw data into a system
3. Processing→Applying formulas, logic or algorithms
4. Output→Generating readable results like reports, visuals
5. Storage→Saving data for future user-friendly

Data Types

- Numeric eg integers, decimals
- Text/String eg letters, words
- Boolean eg True/False
- Date eg Timestamps
- Binary eg 0s and 1s

Cloud Computing Services

Cloud computing refers to delivering computing resouces like storage, servers, software etc over the internet.

Cloud Service Models

- IaaS (Infrastructure as a Service)→Rents servers, storage, networking eg AWS EC2, Microsoft Azure.
- PaaS (Platform as a Service)→Provides tools for app development eg Google App Engine, Heroku.
- SaaS (Software as a Service)→Enables access to software via browsers eg Gmail, Google, Docs, Dropbox.

Cloud services offer scalability, remote access and cost efficiency.

Artificial Intelligence

It is the simulation of human intelligence by machines. It enables systems to learn, reason and solve problems eg chatbots, recommendation engines, autonomous vehicles.

Machine Learning

It is a subset of AI that allows systems to learn from data and improve over time without explicit programming eg supervised, unsupervised and reinforcement learning using tools like TensorFlow, Scikit-learn.

Internet of Things (IoT)

It refers to interconnected devices that collect and exchange data via the internet. eg smart homes, wearable tech, industrial sensors. It allows automation, real-time monitoring and data-driven decisions.

Emerging Technologies

1. Quantum Computing→These computers use qubits for ultra-fast data processing.
2. Edge Computing→Processes data near the source eg IoT devices.
3. Blockkchain→It is a decentralized system ledger for secure transactions like bitcoin
4. 5G Networks→High-speed, low-latency connectivity
5. Green Computing→Eco-friendly tech Practices

Navigating Operating Systems and File Systems

An Operating System is the software that manages hardware and software resouces eg Windows,macOS and Linux.

Desktop Navigation

- Start Menu→It is used to access programs, settings and power options.
- Taskbar→Views open apps, notifications and system tray.
- File Explorer→It browses folders, drives and files.

File System Structure

- Drives→C:\ (main drive), D:\ (secondary or external drives follow this hierarchy)
- Folders→Organize files by categories eg documents, pictures
- Paths→They show location eg \Users\Ashiraf\Documents\BIT

File Operations

- Create→Right click→New→File/folder
- Rename→Right click→Rename
- Move/Copy→Drag and Drop or Ctrl + C/V
- Search→Use file explorer's search bar.

Flowcharts

It is a visual diagram that represnts the steps of an algorithm or process. They help visualize logic before coding.

1. This represents a start/ end.
2. This represents an action or instruction/ process
3. Decision, it is a yes or no/ true or false
4. Flow direction, the direction information is flowing.

Pseudocode

Is a simplified, language-independent way to describe an algorithm using plain English and programming-like structure. Eg

Start

Set total to 0

For each number in list:

    Add number to total

Display total

End

It bridges the gap between human thinking and actual code.

Information Systems

It is a combination of technology, people and processes that work together to manage and deliver information.

Components of Information Systems

1. Hardware→Physical devices like computers, servers, and networking equipments.
2. Software→Programs and applications that process data.
3. Data→Raw facts used to generate meaningful information
4. People→Users, IT, professionals and decision-makers
5. Processes→Procedures and rules for collecting, processing, and distributing data-driven

Importance of IS in Organizations and Society

- Efficiency→It allows automation and reduces manual effort
- Decision Support→Provides accurate, timely information
- Commuincation→Enhances internal and external interactions
- Innovation→Enables new products, services and business models.
- Public Services→Powers e-government, healthcare and transportation

Levels of Information Systems

- Operational level→This handles daily ops like sales, payroll eg TPS
- Managerial level→This level summarize data for middle managers eg MIS
- Strategic level→It supports top-level decision-making eg EIS
- Knowledge→Aid professionals in creating knowledge like CAD tools eg KWS

Types of Information Systems

1. TPS (Transaction Processing Systems)→Records routine operations or transactions of the business eg POS systems.
2. MIS (Management Information System)→Generates reports from TPS data.

3. DSS (Decision Support Systems) → Helps with complex decisions
4. EIS (Executive Information System) → It provides dashboards and summaries for Executives
5. OAS (Office Automation Systems) → Manages documents and communication.
6. ERP (Enterprise Resource Planning) → Integrates all business functions.
7. CRM (Customer Relations Management) → Manages customer interactions

Information System Cycle

- Input - collecting raw data.
- Processing - Converting data into meaningful information
- Storage - Saving data for future use
- Output - Delivering information to users
- Feedback - Using output to refine future inputs

DIKW Pyramid

The DIKW hierarchy explains the transformation of raw data into wisdom.

- Data → Raw facts
- Information → Structured and contextualized data
- Knowledge → Insights derived from information
- Wisdom → judgement and decision-making based on knowledge

Components of Quality Information

Quality Information is one that works to the user's expectations during usage.

1. Accuracy → Free from errors
2. Timeless → Available when needed
3. Relevance → Applicable to the tasks
4. Completeness → All necessary data included
5. Consistency → Uniform across systems
6. Accessibility → Easy to retrieval and use.

Note: System Boundaries defines what is inside the IS (Its components) and external (users, other systems), **Environment** defines the regulations, competitors and market trends that influence the IS.

Databases and DBMS

A database is a structured collection of data that can be easily accessed, managed and updated. A DBMS is software that interacts with users, applictaions and the database itself to manage data efficiently.

Key Functions of a DBMS

- Data storage and retrieval
- Data manipulation and querying
- Security and access controls
- Backup and recovery
- Concurrency and transaction management

Data Storage, Retrieval and Organization

1. Storage → Data is stored in tables for relational databases, non-relational databases use documents, key-value pairs or graphs.
2. Retrieval → Use queries eg SQL to fetch specific data
3. Organization → Data is organized using schemas, which define structure. Relationshps between tables are managed using keys (primary, foreign)

SDLC (System Development Life Cycle)

It is a step-by-step farmework used to plan, create, test and deploy an IS. It ensures that systems meet user requirements, are delivered on time and function reliably.

Components of the SDLC

1. Planning → It is where we define goals, scope, resources and timeline.
2. Feasibility study → Assessing technical, economic, legal, operational and schedule feasibility
3. System Analysis → Gather and analyze user requirements
4. System Design → Create architecture , data models, and interface layouts
5. Development → Write and assemble code based on design
6. Testing → Verify functionality, performance and security
7. Implementation → Deploy system to users and train them
8. Maintenance → Monitor, update, and fix issues post deployment

Feasibility Study

It evaluates whether a proposed system is viable and usable according to the available resources.

Types of feasibility studies

- Technical → Do we have the tools and skills
- Economical → Is it cost-effective
- Legal → Does it comply with laws
- Operational → Will users accept and use it
- Schedule → Can it be completed on time

CSFs (Critical Success Factors)

These are key areas that must go right for a project to succeed.

- Clear objectives and scope
- Strong leadership and stakeholder supports
- Skilled taem and resources
- Effective communication
- User involvement and feedback
- Risk management and contigency planning

JAD (Joint Application Development)

It is a collaborative technique where developers, users and stakeholders work together in structured workshops to define requirements and design solutions.

- Speeds up development
- Improves accouracy of requirements
- Enhances user satisfaction.

System Development Methodologies

1. **Waterfall** → It is a linear approach which follows phase by phase such that it phase can only come after completion of the other. It is used in big projects which require alot of documentation, stable requirements and well-defined projects.

2. **Agile** → It is an iterative, flexible, user-focused method that focuses on producing results earlier and always repeating cycles in order to get the best. It is good for Rapid Development and systems with changing needs.
3. **Spiral** → This methods combines design and prototyping with risk analysis. It is used when projects require very stable systems such that risks are calculated at every stage eg in Banks
4. **RAD (Rapid Application Development)** → It involves fast prototyping and feedback from users because it uses short timelines and user-driven. This method is used to get a working prototype as the final system is worked on.
5. **DevOps (Development and Operations)** → It integrates development and operations incase of systems that always need updates and changing user requirements. It involves Continous Delivery and Continous Deployment (CI/CD)

Factors to consider when choosing a Methodology

- Project size and complexity
- Timeline and budget
- User involvement level
- Risk tolerance
- Team Expertise
- Flexibility and adaptability
- Regulatory or compliance needs

Stakeholders in System Development

- Users
- Project managers
- Developers
- Analysts
- Sponsors
- IT support
- Regulators

System Modeling

It is the practice of creating abstract representations of systems to understand, analyze, and improve their structure and behavior. It helps developers, analysts, and stakeholders visualize how components interact, plus how data flows in the system.

- Models can be **Logical** eg DFDs, ERDs (Entity Relationship Diagrams like Databases) or **Physical** eg network diagrams.
- Used in software engineering, system analysis and project planning.

Data Flow Diagrams

It is a graphical tool used to represent the flow of data within a system. It shows how data enters, moves, processed and exits the system. These systems where introduced by Tom Demarco in 1979 as part of structured analysis, they also help visualize system functionality without focusing on technical details like code or hardware.

Levels of DFD

1. **Level 0 (Context level)** → It represents the entire system as one single process. It shows external entities and data flows to/ from the system. No Internal details, just the system's boundaries. Eg A university system receiving student data and sending results to students and lecturers.
2. **Level 1 (Top-level DFD)** → Breaks the single process into sub-processes. It shows major data stores and more detailed flows, each sub-process is numbered eg Student registration, course management and result processing as separate processes
3. **Level 2 and beyond** → These further decompress level 1 processes into more detailed steps. It is useful for complex systems with many operations, and also processes are numbered hierarchically.

Components of a DFD

1. **Process** → It is either circle or rounded rectangles. Transform data input into output
2. **Data Flow** → These are arrows. It shows movement of data between elements
3. **Data Store** → Open-ended rectangles. They show where data is stored
4. **External Entities** → Square or double square. These are outside source or destinations of data/ sinks.

Rules for constructing a DFD

- Every data flow must connect to atleast one process.

- Processes must have both input and output
- Data stores cannot directly connect to external entities
- Use unique names for each component
- Keep diagrams simple and readable eg use hierarchical levels (level 0, level 1 etc)

Uses of a DFD

- Used in system analysis to understand current workflows
- Used in system design to plan new systems or upgrades
- It helps in communication by aligning developers, clients and stakeholders
- Helps in troubleshooting by identifying bottlenecks or inefficiencies

DFD Inventions and Notations

1. Gane & Sarson → Rounded rectangles for processes and open sided rectangles for data stores
2. Yourdon → Circles for processes and open rectangles for data stores

Note: All DFDs use squares as their external entities

Defintion and History of the Internet

The internet is a vast network of interconnected computers and servers that enables global communication and data exchange. It began in 1969 as ARPANET, a US Dept of Defense project to connect research institutions. By the 1980s, protocols like TCP/IP standardized communication and in 1990, the internet became publicly accessible.

World Wide Web (WWW)

It is a system of interlinked hypertext documents accessed via the internet. It was invented by Tim Berners-Lee in 1989 to help scientists share information. The web uses HTTP and URLs to connect pages and resources.

Fundamentals of the WWW

1. Web pages → These are documents structured by html.
2. Hyperlinks → Connections in the www that allow a seamless access of different webpages
3. Web browsers → Softwares that display and relay web pages
4. Search engines → These index and search for requested web pages by the browsers
5. Web servers → They host websites on the internet.

Web Development Tools

- Html/CSS/JavaScript → Core languages
- VS code / Sublime Text → Code editors
- Bootstrap / Tailwind → CSS frameworks
- Git / GitHub → Version controls
- Figma / Adobe XD → UI/UX design
- CMS platforms → WordPress, Joomia

URL (Uniform Resource Locator)

It specifies address of a resource eg **https://ghostr.tech/portfolio**

Parts → Protocol (https://), Domain Name (ghostr.tech), Path (/portfolio).

URI (Uniform Resource Identifier)

It is a broader term that includes URLs and URNs (names without location), allows a user to locate and identify an item on the internet by specifying the location.

Requirements to Create and Publish a Website

1. Domain Name → Register via providers like GoDaddy or Namecheap.
2. Web Hosting → Choose a web server to host your services
3. Website files → Build using HTML/CSS/JavaScript or CMS
4. FTP → Upload files to a server
5. SSL Certificate → For secure HTTPS access
6. Testing and Launch → Check responsiveness, links and perfomance.

Client-Server Model

It is a distributed computing architecture where clients like browsers send requests for services or data and servers like web servers process and respond to those requests.

Web Servers

These store, process and deliver web pages to clients over the internet using protocols like http/s eg Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed, Google Web Server, AWS.

MIME (Multipurpose Internet Mail Extensions)

It is a standard that indicates the nature and format of a file or data being transmitted over the internet. It is used in HTTP headers to tell browsers how to handle content.

MIME Types

1. Text → text/plain, text/html, text/css
2. Image → image/jpeg, image/png, image/gif
3. Audio / video → audio/mpeg, video/mp4
4. Application → application/json, application/pdf, application/zip
5. Multipart → multipart/form-data (used in file uploads)

Structure of a Basic Web Page

Every HTML page starts with a standard structure

- <!DOCTYPE html>
- <html>
  - <head>
    - <title>My Website</title>
    - <link rel="stylesheet" href="styles.css">
  - </head>
  - <body>
    - <header>Welcome to GhostR site</header>
    - <nav><a href="about.html">About</a></nav>
    - <main><h1>Home Page</h1></main>
    - <footer>© 2025 ASHIRAF</footer>
  - </body>
- </html>

Elements of a WebPage

1. Header → Branding, navigation
2. Main → Core content

3. Sidebar → Optional tools or links
4. Footer → Contact info, copyright
5. Forms → User input

HTML Basic Tags

1. **Text tags:**

```
<h1>
<p>
<strong>
<em>
<bold>
```

2. **Lists:**

```
<ul><li>Well this is an unordered list.</li></ul>
<ol><li>This is an ordered list.</li><ol>
```

3. **Tables:**

```
<table>
    <tr><th>Name</th><th>Age</th></tr>
    <tr><td>GhostR</td><td>20</td><tr>
</table>
```

4. **Forms:**

```
<form action="submit.php" method="post">
    <label>Name: <input type="text" name="name"></label><br>
    <input type="radio" name="gender" value="male">Male
    <input type="radio" name="gender" value="female">Female<br>
    <select>
        <option value="BIT">BIT</option>
        <option value="DIT">DIT</option>
    </select><br>
```

```
</form>
```

Semantic Elements

These are elements that improve accessibility, SEO and code clarity.

- <header>
- <nav>
- <main>
- <section>
- <article>
- <aside>
- <footer>

GhostR